Rational Number Project

Fraction Operations and Initial Decimal Ideas **Lesson 9: Overview**	Materials
Students create a model for decimals using a 10 x 10 grid to show tenths and hundredths. They record amounts in words and fractions with denominators of 10 and 100.	• Rulers for students • Student Pages A- B • Orange and yellow colored crayons, pencils or markers for each student

Teaching Actions

Warm Up

Order the fractions from smallest to largest. Be ready to explain your reasoning.

$$\frac{5}{10} \quad \frac{87}{100} \quad \frac{49}{100} \quad \frac{5}{100}$$

Large Group Introduction

1. In a large group setting, project the square (Student Page A) for all to see. Students should have their own copies.

2. Explain: Using the notches at the top and bottom of the square, use your ruler to draw lines down the square.

3. Ask: What did you do to the square? If the square is our whole unit, and I color three of the ten parts orange, what fraction of the whole square is orange? What amount is not shaded? (At this point, emphasize verbal mode; don't record with symbols).

Comments

Create a 10 x 10 grid on your Smart Board or create a large poster size 10 x 10 grid on card stock, laminated. Cut out from poster board orange strips to represent 10ths and yellow squares to represent 100ths. Have this available for each decimal lesson.

Color is important as students construct mental images for tenths and hundredths. Students will remember not only the size of tenths and hundredths in relation to the 10 x 10 grid but also the color. That is why we recommend that students use orange and yellow crayons, pencils or markers to do the shading activities in the decimal part of the module.

An important skill for students to develop is the ability to compose and decompose decimals. This skill involves being able to interpret a decimal flexibly. For example .34 can be viewed as 3 tenths and 4 hundredths. But also students should be able to reconfigure or compose numbers if they are given 3 tenths and 4 hundredths to .34. Students who can do this are able to order decimals conceptually as well as build meaning for adding and subtracting decimals in a meaningful way.

Teaching Actions

4. Return to the grid and draw lines from left to right.
    - Ask: How has the whole changed?
    - How many small squares in all? If I color 5 small squares yellow, what fraction of the whole square is yellow?
    - How many small squares in 1-tenth of the square? What can you say about 10-hundredths and 1-tenth?
    - Name the amount shaded in orange and yellow combined (Encourage a variety of ways to do this: 3/10 + 5/100; 35 out of 100; 3 orange strips and 5 small yellow squares; if all in yellow it would be 35 yellows; 35/100).

5. Show the grid with 35 small squares in yellow. Ask students how this model is similar and different to the one with 3 orange strips and 5 yellow squares.

6. Ask: Where are the 3-tenths in the model all in yellow? (Outline that amount with orange). Name the total amount shaded in a variety of ways and note how they are the same as the model in orange and yellow.

Small Group/Partner Work

7. Ask students to use their crayons to show the amounts found in the table below in two different ways. Use Student Page B.

Show this amount in two different ways on two different 10 x 10 grids	Potential ways of describing the pictures
34 out of 100 equal parts	$\frac{34}{100}$; $\frac{3}{10}$ and $\frac{4}{100}$; 34-hundredths; 34 yellow squares; 3 orange strips and 4 yellow squares more
2 out of 10 equal parts and 6 out of 100 equal parts	$\frac{2}{10}$ and $\frac{6}{100}$; $\frac{26}{100}$; 26-hundredths; 2 orange strips and 6 yellow squares; 26 yellow squares
55 hundredths	$\frac{55}{100}$; 55 yellow squares; 5 orange strips and 5 yellow squares; $\frac{5}{10}$ and $\frac{5}{100}$

Comments

You can see how composing and decomposing helps students order decimals in this example. Consider how the grids and the colors support these students' understanding of decimal size.

T: Which is bigger .75 or.9? What do you picture when you see the 75-hundredths?
S: I picture 7 oranges and 5 yellow squares.
T: And the tenths?
S: I see 9 oranges.
T: So then which is bigger?
S: 9-tenths.

Consider one more example of student's thinking based on the 10 x 10 grid and the ability to be flexible on how he interprets a decimal amount:

> 2.3 - .05. I imagine this one with a grid so 2 full grids and then there's 3 tenths so you minus 5 of the hundredths because 3 tenths is the same as 30 hundredths. Cross out 5 hundredths which leaves you with 25. Two and 25-hundredths.

Teaching Actions

Wrap Up

8. When students are done, share their pictures and record different ways to describe each representation. (See the table on the previous page for ideas). Encourage a variety of ways to describe the amounts on the board. You want students to be flexible in how they interpret the each visual. Record what students say on the board.

9. Explain that in the next lesson they will record amounts on the 10 x 10 grid using decimal notation. (For this lesson, use verbal descriptions based on color, fraction language (3-tenths and 5-hundredths; 35-hundredths) and fraction symbols (3/10 + 5/100; 35/100).

Comments

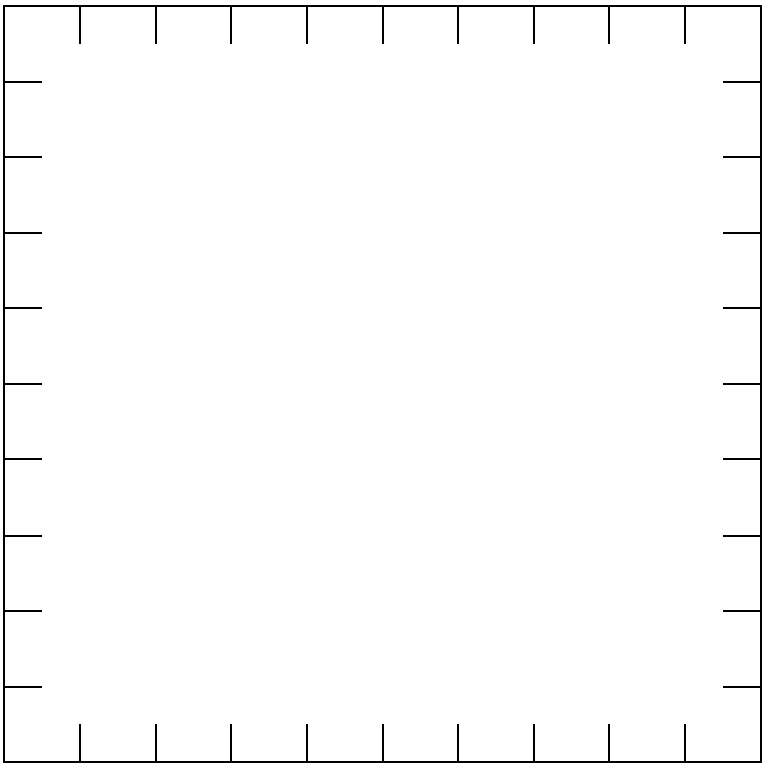
Translations:

- Concrete to verbal
- Concrete to verbal to symbols
- Concrete to verbal to pictures to symbols
- Symbols to pictures to symbols

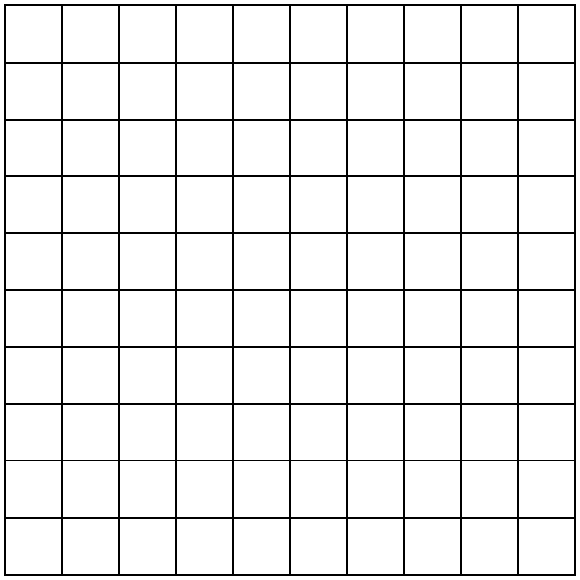
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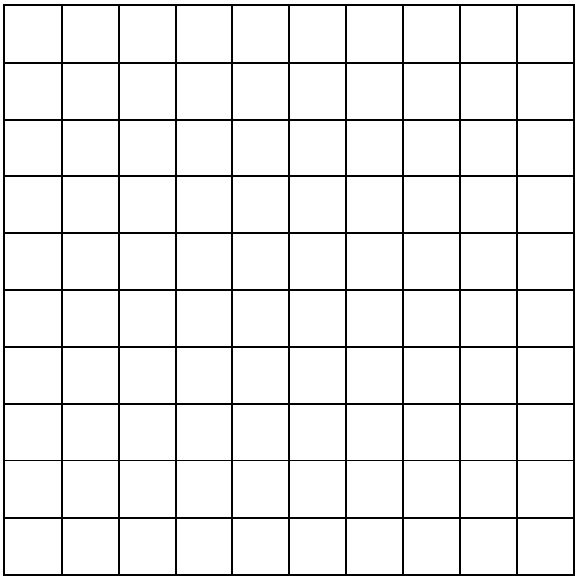
$$\frac{5}{10} \quad \frac{87}{100} \quad \frac{49}{100} \quad \frac{5}{100}$$

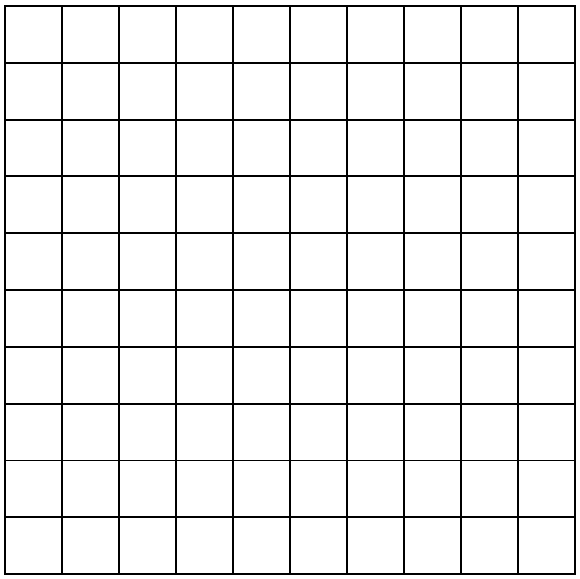
Name_____ Lesson 9/Student Page A

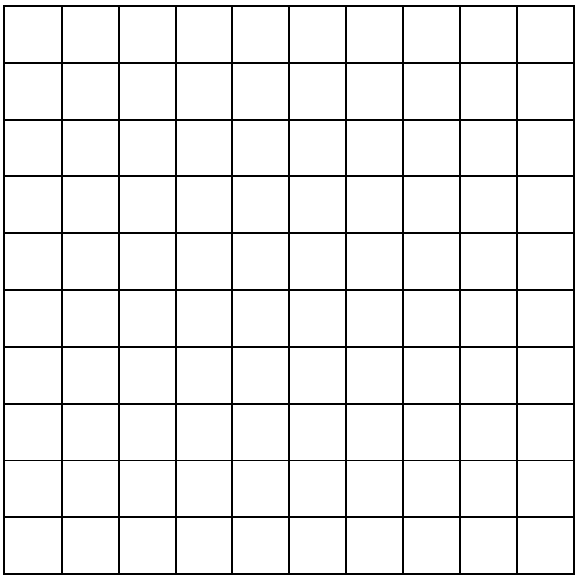


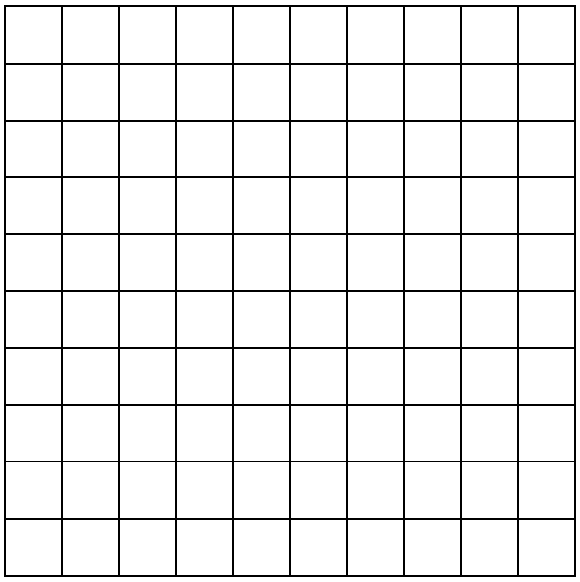
Name_____ Lesson 9 / Student Page B

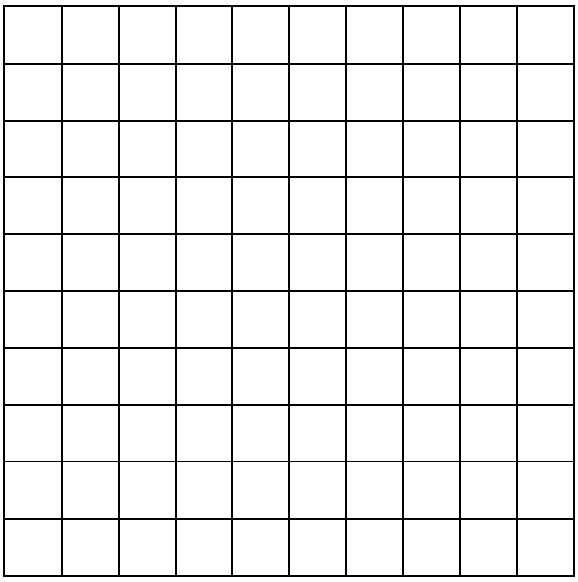












Post Lesson Reflection

Lesson_____

1) Number of class periods allocated to this lesson: _____

2) Student Pages used: _____

3) Adaptations made to lesson: (For example: added extra examples, eliminated certain problems, changed fractions used)

4) Adaptations made on Student Pages:

5) To improve the lesson I suggest: